

# DYNAMO: Dynamic Skill-Tool Evolution for Vision-Language Agents

Anonymous ACL submission

## Abstract

Improving vision-language models (VLMs) on visual reasoning typically requires retraining or hand-designed prompts and tools. We present DYNAMO, a training-free framework in which a frozen VLM agent inspects its own correct and incorrect attempts on a small training subset and evolves two complementary capabilities: reusable reasoning skills for cognitive bottlenecks, and executable visual tools—gated by learned applicability—for perceptual ones, accumulating them in a persistent library without weight updates. Across four visual reasoning benchmarks and six VLM backbones, DYNAMO improves direct inference in 21 of 23 settings (avg. +4.8 acc). The same framework extends to the tool-mastery setting on GTA, where mastery profiling lifts step-level tool selection across all tested backbones. Against task-specific RL (VTool-R1, DeepEyes), DYNAMO matches or approaches RL accuracy at a fraction of the compute (60–94% gap recovery on chart/table; 91.1 vs. 90.1 on V\*-7B) and combines additively with RL when available.

## 1 Introduction

Vision-language models (VLMs) have improved rapidly, yet adapting a frozen VLM to a new visual-reasoning task family still typically requires manually curated SFT data or a hand-designed RL pipeline—per task and per backbone. We ask whether the agent can build its own task-family-specific capability set instead, by inspecting its own behaviour on a small training subset and without any weight updates.

We instantiate this idea in DYNAMO, a training-free framework that evolves two complementary capability types and dynamically equips them at inference. **Skills** are structured Markdown SOPs for *cognitive* bottlenecks, where the visual evidence is available but the agent’s reasoning procedure is weak. **Tools** are short Python programs for *perceptual* bottlenecks, where the agent needs a trans-

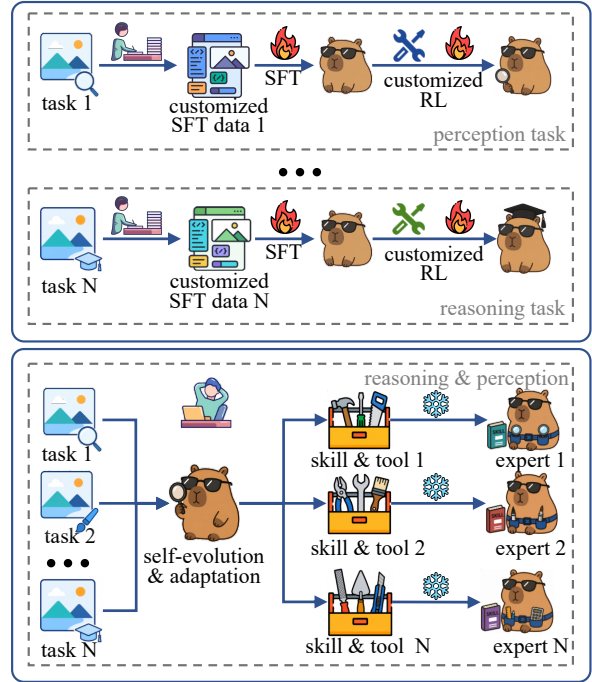


Figure 1: **Labor-free task adaptation via skill-tool evolution.** Per-task VLM adaptation typically requires manually curated SFT data with a hand-designed RL pipeline for every new task family. DYNAMO instead lets a frozen VLM agent evolve its own skill and tool set per task family without human supervision, and dynamically equips the matching set when a case from that family arrives.

formed view—a crop, zoom, contrast adjustment, or rerendering—before the evidence becomes legible. On each iteration over a sampled sub-training set, DYNAMO diagnoses both the correct and the incorrect attempts using their images and reasoning traces, proposes multiple candidate skill+tool combinations, and promotes the highest-validation-accuracy candidate into a persistent library. A mastery phase further learns when each tool is safe to invoke, so that promoted capabilities are deployed selectively rather than indiscriminately.

We evaluate three questions. (I) Does DY-

**NAMO work across diverse benchmarks and backbones?** On ChartQA, MathVista, HRBench, and V\* across six VLM backbones, skill-tool co-evolution improves direct inference in 21 of 23 model–benchmark settings, averaging +4.8 accuracy points. **(II) Does the framework extend to a tool-mastery setting with a curated tool set?** On GTA, mastery profiling lifts step-level tool selection, argument, and instruction-following accuracy across all tested backbones. **(III) Can DYNAMO match task-specific RL across different task families?** Across three task families—chart, table, and high-resolution perception—we compare DYNAMO to a representative RL method on each (VTool-R1 for chart/table; DeepEyes for perception). DYNAMO matches or approaches RL accuracy at a fraction of the compute (60–94% gap recovery on chart/table; 91.1 vs. 90.1 on V\*-7B perception), and combines additively with RL when both are available.

## Contributions.

- (1) A **problem formulation** that recasts per-task VLM adaptation as labor-free capability evolution: a frozen agent builds its own task-family-specific skill and tool set from a small training subset, replacing manual SFT data curation and task-specific RL pipelines.
- (2) A **multi-candidate evolution loop with mastery** that diagnoses both correct and incorrect attempts, generates candidate skill+tool combinations, selects the best by validation accuracy, and learns per-tool applicability boundaries.
- (3) **Empirical evidence** across four visual-reasoning benchmarks, six VLM backbones, the GTA tool-mastery benchmark, and three task families (chart, table, perception) each compared against its representative RL method, showing that capability evolution is a viable training-free alternative—and add-on—to task-specific RL.

## 2 Related Work

**Self-improving agents.** A growing line of work lets LLM agents improve from experience without gradient updates: Reflexion (Shinn et al., 2023) keeps a verbal reflection buffer that resets between episodes; Voyager (Wang et al., 2024) and ExpeL (Zhao et al., 2024) accumulate persistent skill libraries in game and tool-use domains; AutoManual (Chen et al., 2024) and EvolveR (Anonymous,

2025a) refine instruction manuals or reasoning strategies via self-play. All operate in text, code, or game domains, never generating executable visual processing nor deciding whether a bottleneck is cognitive or perceptual. DYNAMO extends the persistent-library paradigm into the visual domain with a multimodal diagnostic that text-only frameworks cannot perform.

**Tool creation for language models.** A complementary line generates new tools on demand: CREATOR (Qian et al., 2023) prompts an LLM to write Python helpers for problems it cannot solve directly; LATM (Cai et al., 2023) and ToolLLM (Qin et al., 2024) scale this to large API collections. These methods generate tools in textual or mathematical domains and do not profile when a generated tool succeeds or fails—a gap that matters for visual tools, where the same operation can help on most images but hurt on specific patterns (e.g. a sharpening filter degrading low-contrast charts). DYNAMO’s mastery phase profiles each tool on a held-out case set to characterise its applicability before promotion.

**Visual reasoning with tools.** Several recent systems equip VLMs with visual processing tools. VTool-R1 (Anonymous, 2025d), DeepEyes (Zheng et al., 2025), Adaptive-CoF (Zhang et al., 2025), PixelReasoner (Anonymous, 2025b), and V-Thinker (Anonymous, 2025c) all rely on SFT and/or RL with curated trajectories to teach a VLM when and how to invoke a fixed tool inventory. ZoomEye (Shen et al., 2024) and ReFocus (Fu et al., 2025) are training-free but assume a fixed zoom/search algorithm or visual-editing interface. DYNAMO instead generates and validates the skills and tools themselves from the agent’s own behaviour on a small training subset, with no weight updates and no predefined tool inventory.

**Tool-augmented VLMs with predefined tool sets.** VisProg (Gupta and Kembhavi, 2023), ViperGPT (Surís et al., 2023), and Chameleon (Lu et al., 2023) compose hand-curated tool libraries (detectors, OCR, arithmetic modules) via program synthesis. They handle broad compositional visual tasks effectively but cannot recover when the required tool is missing from the human-designed inventory. DYNAMO fills this gap by generating new tools when the current set is insufficient, bridging fixed-tool composition and training-based tool learning.

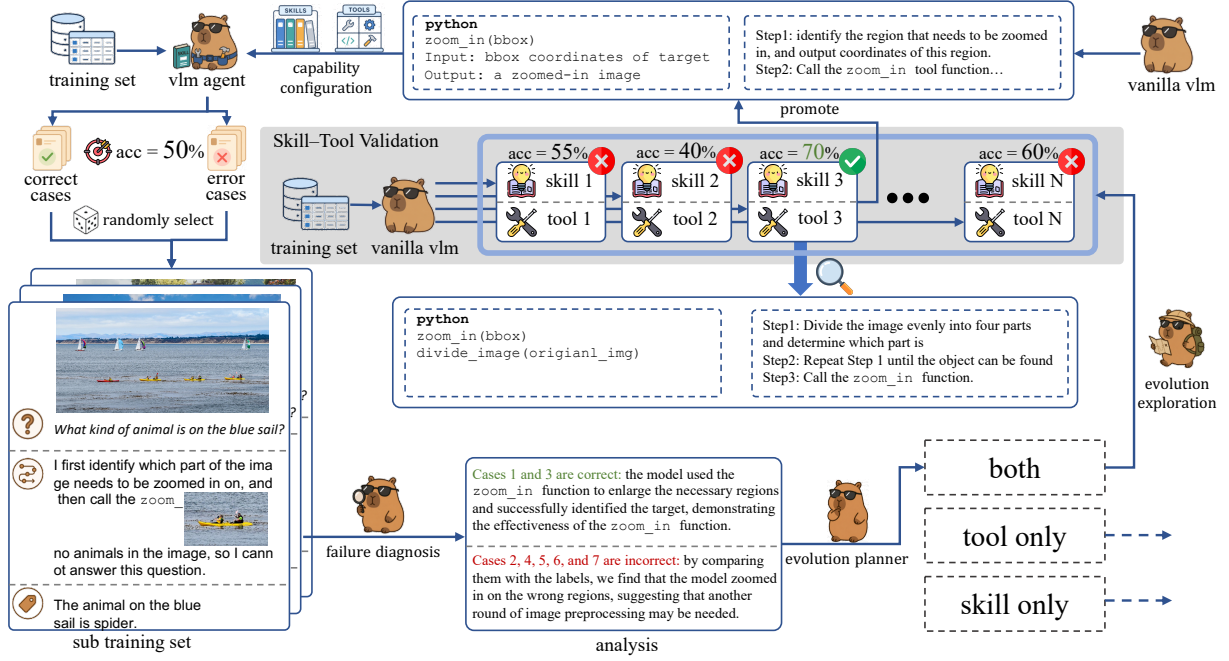


Figure 2: **DYNAMO evolution loop.** On a sampled sub-training set, DYNAMO diagnoses both correct and incorrect attempts, explores candidate skill+tool capabilities, and promotes the best by validation into a persistent library.

### 3 Method

#### 3.1 Problem Formulation

Let  $\mathcal{D}_{\text{train}} = \{(x_i, y_i, \mathbf{v}_i)\}_{i=1}^k$  denote a small training subset of  $k$  cases, where  $x_i$  is the question,  $y_i$  the ground-truth answer, and  $\mathbf{v}_i$  the associated visual input (one or more images). Let  $\pi_\theta$  be a frozen VLM backbone (no gradient updates permitted). We define a capability set  $\mathcal{C} = \{\mathcal{S}, \mathcal{T}\}$ , where  $\mathcal{S}$  is a library of *skills* (structured reasoning SOPs) and  $\mathcal{T}$  is a library of *tools* (executable Python programs). An agent  $f_{\mathcal{C}}$  operates by retrieving relevant capabilities from  $\mathcal{C}$ , applying them during problem solving, and invoking  $\pi_\theta$  for reasoning. Our goal is to learn  $\mathcal{C}$  from  $\mathcal{D}_{\text{train}}$  to maximise performance on a held-out validation set  $\mathcal{D}_{\text{val}}$ , subject to the constraint  $\nabla_\theta = 0$  (no weight updates).

#### 3.2 Evolution Loop

Each evolution iteration over  $\mathcal{D}_{\text{train}}$  has three phases (Figure 2).

**Diagnose.** The agent solves  $\mathcal{D}_{\text{train}}$  with the current  $\mathcal{C}$ , then samples a sub-training set  $\mathcal{D}_{\text{sub}} \subseteq \mathcal{D}_{\text{train}}$  containing both correct and incorrect attempts. The ANALYZERDECIDER inspects  $\mathcal{D}_{\text{sub}}$ —the questions, original images  $\mathbf{v}_i$ , reasoning traces  $\tau_i$ , and any intermediate tool outputs—and emits a root-cause analysis grounded in visual evidence, an identified bottleneck type (*cognitive* when the visual evidence is available but the reasoning pro-

cedure is weak, or *perceptual* when the agent needs a transformed visual input), and an action  $a \in \{\text{skill}, \text{tool}, \text{both}\}$ . Visual input here is essential: deciding whether a crop is misaligned, whether labels are legible, or whether a processed image introduced artefacts requires inspecting the image, not just the reasoning trace.

**Explore.** Conditioned on the diagnosis, the GENERATOR proposes  $M$  candidate skill+tool combinations, each addressing the diagnosed bottleneck with a different strategy or implementation.

**Validate and promote.** Each candidate is evaluated on  $\mathcal{D}_{\text{train}}$ , and the combination with the highest accuracy is added to  $\mathcal{C}$ . Selection by accuracy on  $\mathcal{D}_{\text{train}}$  subsumes both an *origin* requirement (the new capability must improve over the previous  $f_{\mathcal{C}}$  on the cases it targets) and a *regression* requirement (it must not degrade currently-correct cases). When the action is both, the paired skill additionally serves as the tool’s mastery SOP, gating tool invocation at inference (Section 3.4). The full procedure runs for  $N$  iterations (default  $N=3$ ); pseudocode is given in Appendix A (Algorithm 1).

#### 3.3 Skills

A *skill* is a structured Markdown document with four fields: **When-to-Use** (a trigger predicate over question and image type), **Strategy** (a numbered step-by-step SOP), **Common Pitfalls**, and **Worked**

**Example.** The Generator writes a skill conditioned on the AnalyzerDecider’s root-cause analysis; if a skill covering the same problem class already exists in  $\mathcal{S}$ , new insights are merged into it rather than creating a duplicate, preventing library bloat. At inference, the Solver retrieves the top- $K$  skills by BM25 similarity to the question and appends them to the system prompt.

### 3.4 Tools and Mastery

A *tool* is a Python function ( $\leq 150$  lines) that takes one or more image paths and returns processed images or extracted data, using standard CV libraries (OpenCV, PIL, NumPy). The Generator writes a tool conditioned on a perceptual bottleneck diagnosis; representative examples produced in our experiments include chart re-rendering with enlarged axis labels, saliency-guided sub-region extraction, and contrast enhancement for low-quality documents.

**Mastery as a paired skill.** When the AnalyzerDecider’s action is both, the Generator emits a skill alongside the tool, and this paired skill *is* the tool’s mastery SOP. Its When-to-Use predicate specifies the input patterns on which the tool helps, its Common Pitfalls flag the patterns on which it hurts, and its Strategy instructs the Solver how to invoke the tool and consume its output. Because tool invocation is gated by retrieving the paired skill, deployment is selective by construction rather than indiscriminate—a property that distinguishes DYNAMO from prior tool-creation methods (Qian et al., 2023; Cai et al., 2023) that expose generated tools without learned applicability boundaries.

**External tool sets (Mode B).** When a curated tool set  $\mathcal{T}_0 \neq \emptyset$  is already provided, DYNAMO can operate in **Mode B**: skip tool generation and instead synthesise a mastery skill for each provided tool  $t_j \in \mathcal{T}_0$ , learning when to invoke it from the agent’s own behaviour on  $\mathcal{D}_{\text{train}}$ . Mode B requires no code generation, making it practical when tool quality is already high but deployment strategy is unknown. Experiment II (Section 4.3) evaluates this mode.

### 3.5 Online Adaptation to Distribution Shift

In real deployments, DYNAMO runs against a streaming query distribution that can shift over time—a stream may begin with high-resolution perception cases and later start including MathVista-style reasoning cases, or vice versa. We design DYNAMO to detect such shifts and *update* its skills

and tools online, without weight updates and without the practitioner having to anticipate the shift or prepare a per-family training set.

DYNAMO continuously samples a sub-training set from the most recent queries and runs the evolution loop of Section 3.2 on them, accumulating skills and tools into the library  $\mathcal{C}$ . A rolling-window monitor tracks the dominant task family in recent queries from their question and image features; when the dominant family changes, the next evolution iteration is triggered on the new sub-stream and produces updated skills and tools that handle the new distribution. The library is thus kept in sync with the current query distribution without any human intervention. We treat this autonomous online adaptation as one of the method’s contributions; Experiment I (distribution-shift adaptation, Section 4.2) evaluates it against a static baseline that never refreshes and an oracle whose full capability set is available from the start.

## 4 Experiments

### 4.1 Experimental Setup

**Benchmarks.** For the evolution-from-scratch setting (Exp I) we use four visual reasoning benchmarks: **ChartQA** (Masry et al., 2022) (multi-step numerical reasoning over charts; 2,500 test questions, relaxed accuracy with 5% tolerance), **MathVista** (Lu et al., 2024) (math reasoning grounded in figures and plots; 1,000 testmini questions, accuracy), **HRBench** (Anonymous, 2024b) (perception on  $\geq 4K$ -resolution images; accuracy), and **V\*** (Wu and Xie, 2023) (visual search for an object’s property;  $\approx 500$  questions, accuracy); the first two stress *cognitive* bottlenecks while the latter two stress *perceptual* ones. For the tool-mastery setting (Exp II) we use **GTA** (Anonymous, 2024a), a general tool-agent benchmark that exposes a curated tool set and reports step-level metrics for tool selection, argument, instruction-following, and final-answer accuracy. For the RL comparison (Exp III) we follow the VTool-R1 protocol (Anonymous, 2025d) on its **Chart** and **Table** reasoning splits and reuse **V\*** and **HRBench4K** as the high-resolution splits of the DeepEyes protocol (Zheng et al., 2025).

**Foundation models.** Experiment I evaluates six proprietary and open-weight VLMs: **GPT-4o**, **o4-mini**, **GPT-5.4**, **Doubao-Seed-2.0**, **Qwen3.5-72B**, and **Gemini-3.1-Pro-Preview**. Experiment II evaluates **GPT-4o**, **GPT-5.4**, and **Gemini3Flash** on



the GTA benchmark. We additionally report a controlled Doubao-Seed-2.0-Pro GTA analysis to inspect how provided-tool skills change with tool abstraction and task environment. Experiment III uses the **Qwen2.5-VL** family (3B, 7B, 32B) to align with the VTool-R1 evaluation protocol. All VLM weights are *frozen* throughout; no gradient updates are performed.

**Evolution protocol.** For each benchmark, we sample a training subset of  $k=200$  cases from the official training split and run the DYNAMO evolution loop for  $N=3$  iterations. At the end of training, the capability set  $\mathcal{C}$  is frozen and evaluated on the held-out validation / test split. We report the average over three independent random seeds (different  $k=200$  subsets) to account for sampling variance where repeated runs are available.

**Baselines.** For Experiment I, we compare four evolution configurations that isolate the contribution of each component: **None (Base Agent)** invokes the frozen VLM with no evolved capabilities — the zero-shot baseline; **Skill Only** runs the evolution loop but restricts the Generator to producing reasoning skills (no tool generation); **Tool Only** restricts generation to executable visual tools (no skill generation); **Skill + Tool** is the full DYNAMO system (co-evolution of both capability types).

**Evaluation metrics.** All primary metrics are task-level accuracy on the held-out split. Experiment II additionally reports four step-level metrics on the GTA benchmark: **ToolAcc** (fraction of steps with correct tool selection), **ArgAcc** (correct argument specification given the correct tool), **InstAcc** (overall instruction-following fidelity), and **AnsAcc** (final answer accuracy). For the supplementary controlled GTA analysis, we also report tool usage rate, average number of tool calls, and dominant tool-call chains. Experiment III follows the VTool-R1 evaluation protocol (Anonymous, 2025d) on Chart and Table reasoning splits, and uses DeepEyes (Zheng et al., 2025) as the RL reference for high-resolution  $V^*$  and HRBench4K visual-search tasks.

## 4.2 Experiment I: Autonomous Evolution from Scratch

**Motivation.** The central claim of DYNAMO is that a VLM agent can bootstrap useful capabilities from failure on a small training subset, without any weight updates. Experiment I tests this claim

in the most challenging setting: the capability library is initialised to empty ( $\mathcal{S}=\emptyset$ ,  $\mathcal{T}=\emptyset$ ), and the system must generate, validate, and accumulate all capabilities autonomously. By evaluating six foundation models, with some ablation cells still pending, we assess whether the observed gains are tied to a single backbone or appear across model families.

### Qualitative inspection of evolved capabilities.

In addition to aggregate accuracy, we inspect three representative forms of evolved capability: (i) a ReFocus-style chart/table editing case, where a generated tool marks or masks visual evidence before reasoning; (ii) a ZoomEye-style high-resolution search case, where a generated tool performs coarse-to-fine region inspection; and (iii) an interleaved visual skill, where the retrieved skill contains both a textual SOP and visual evidence from a prior failure. These case studies are qualitative: they support the claim that DYNAMO can generate capability structures resembling useful prior motifs, not that it reproduces the full algorithms of ReFocus (Fu et al., 2025) or ZoomEye (Shen et al., 2024). Appendix D gives the detailed write-up and marks the exact fields to fill with the corresponding generated artifacts.

**Multi-model comparison.** Table 1 reports the completed model–strategy entries with temporary standard-deviation placeholders for layout inspection; the original no-std table source is kept hidden below unchanged. Blank cells indicate runs whose results have not yet been inserted and are excluded from aggregate statements. Comparing the full Skill + Tool row with the base agent, DYNAMO improves direct inference in 21 of 23 completed model-benchmark cells, with an average gain of 4.8 accuracy points. The largest gains occur on Gemini-3.1-Pro-Preview / ChartQA (+14.6), o4-mini / HRBench (+11.6), GPT-5.4 / HRBench (+10.3), GPT-5.4 /  $V^*$  (+9.9), and Doubao-Seed-2.0 /  $V^*$  (+8.6).

The table also shows why the framework should not be interpreted as a monotonic “add more capabilities” recipe. Full co-evolution is not the best entry in every cell: for GPT-4o on MathVista and ChartQA, Skill Only is slightly higher than Skill + Tool; for o4-mini on ChartQA and Qwen3.5-72B on  $V^*$ , the base agent remains higher than the full system. Tool Only can also be harmful, most visibly for o4-mini on HRBench. These exceptions are important: they motivate validation

| Model           | Evolution Strategy | HRBench       | MathVista     | V*            | ChartQA       |
|-----------------|--------------------|---------------|---------------|---------------|---------------|
| GPT-4o          | None (Base Agent)  | 0.6386        | 0.6711        | 0.5828        | 0.7552        |
|                 | Skill Only         | 0.6733±0.0121 | 0.7122±0.0038 | 0.6424±0.0066 | 0.7785±0.0026 |
|                 | Tool Only          | 0.6452±0.0044 | 0.6900±0.0029 | 0.6225±0.0066 | 0.7722±0.0029 |
|                 | Skill + Tool       | 0.6643±0.0052 | 0.7011±0.0067 | 0.6313±0.0101 | 0.7799±0.0045 |
| o4-mini         | None (Base Agent)  | 0.73          | 0.753333      | 0.72847       | 0.7833        |
|                 | Skill Only         | 0.7724±0.0058 | 0.7470±0.0107 | 0.7616±0.0066 | 0.7866±0.0151 |
|                 | Tool Only          | 0.7457±0.1670 | 0.7356±0.0328 | 0.8387±0.0629 | 0.7920±0.0213 |
|                 | Skill + Tool       | 0.8571±0.0099 | 0.7559±0.0055 | 0.7991±0.0399 | 0.8007±0.0081 |
| gpt5.4          | None (Base Agent)  | 0.7471        | 0.7589        | 0.7748        | 0.7974        |
|                 | Skill Only         | 0.7805±0.0036 | 0.7752±0.0039 | 0.7682±0.0199 | 0.8090±0.0022 |
|                 | Tool Only          | 0.8367±0.0346 | 0.7663±0.0061 | 0.8477±0.0132 | 0.8108±0.0052 |
|                 | Skill + Tool       | 0.8538±0.0036 | 0.7704±0.0134 | 0.8411±0.0369 | 0.8276±0.0278 |
| Doubao-Seed-2.0 | None (Base Agent)  | 0.8214        | 0.8544        | 0.8675        | 0.8152        |
|                 | Skill Only         | 0.8600±0.0049 | 0.8556±0.0109 | 0.8609±0.0115 | 0.8198±0.0031 |
|                 | Tool Only          | 0.9005±0.0022 | 0.8659±0.0061 | 0.9448±0.0038 | 0.8156±0.0068 |
|                 | Skill + Tool       | 0.8990±0.0064 | 0.8681±0.0105 | 0.9558±0.0038 | 0.8347±0.0295 |
| Qwen3.5-27B     | None (Base Agent)  | 0.8171        | 0.8088        | 0.88079       | 0.8114        |
|                 | Skill Only         | 0.8721±0.0030 | 0.8152±0.0080 | 0.8675±0.0066 | 0.8097±0.0090 |
|                 | Tool Only          | 0.8986±0.0000 | 0.8241±0.0034 | 0.9514±0.0101 | 0.8115±0.0041 |
|                 | Skill + Tool       | 0.8986±0.0049 | 0.8233±0.0011 | 0.9603±0.0175 | 0.8158±0.0026 |

Table 1: **Exp I: Autonomous evolution from scratch.** Accuracy for each evolution strategy across six foundation-model blocks and four benchmarks, all starting from an empty capability set. *None* is the zero-shot VLM baseline. *Skill Only* / *Tool Only* ablate one capability type. *Skill + Tool* is the full DYNAMO system. Blank cells indicate runs whose results have not yet been inserted.

and mastery rather than unconditional tool use, and they suggest that the benefit of a generated capability depends on both the benchmark and the backbone’s ability to follow the resulting SOP or tool policy.

A cautious pattern is nevertheless visible. Skill-only evolution is often competitive on ChartQA and MathVista, where improvements can plausibly come from better decomposition and calculation procedures. Tool-related variants tend to help more on HRBench and V\*, where sub-region inspection and visual search are more central. We treat this as evidence that the cognitive/perceptual distinction is a useful design rule for capability generation, not as a standalone proof that every error in these benchmarks has that structure.

**Distribution-shift adaptation.** The previous comparison evaluates each benchmark independently. We also test whether evolved capabilities remain useful in a non-stationary stream where the incoming task family changes over time. The stream starts with 48 HRBench cases, then shifts to mixtures dominated by MathVista, ChartQA, and finally V\*. This setting is a controlled replay over completed per-case model outputs: it isolates the

adaptation policy from the cost and stochasticity of regenerating skills and tools online.

We compare four policies. *Direct VLM* uses no persistent capability pack. *Static HRBench-only* starts with the HRBench pack and never adapts. *Online adaptive* also starts with HRBench, but detects new benchmark families in a rolling window and activates the matching capability pack after a small latency. *Oracle all packs* is an upper bound that has all task-family packs available from the beginning. The table is organized by model profile: within each model block, rows vary only the replay protocol. *Capability-relevant* denotes a controlled stream enriched with cases whose outcome depends on whether the matching evolved capability pack is available; *Stress* further concentrates on cases where the relevant pack corrects failures of weaker policies.

Table 2 shows that online adaptation is consistently closer to the oracle than to the static policy. On the main GPT-5.4 capability-relevant stream, online adaptation reaches  $0.889\pm0.011$  accuracy, compared with  $0.654\pm0.016$  for static HRBench-only and  $0.909\pm0.009$  for the oracle. The stress stream makes the same point more sharply: when

| Replay protocol                   | Direct      | Static HR   | Online adapt | Oracle      |
|-----------------------------------|-------------|-------------|--------------|-------------|
| <i>GPT-4o skill+tool</i>          |             |             |              |             |
| Capability-relevant               | 0.418±0.020 | 0.558±0.023 | 0.789±0.025  | 0.818±0.022 |
| Stress                            | 0.068±0.009 | 0.347±0.011 | 0.912±0.008  | 0.961±0.010 |
| Natural                           | 0.664±0.032 | 0.673±0.030 | 0.702±0.028  | 0.705±0.028 |
| <i>o4mini skill+tool</i>          |             |             |              |             |
| Capability-relevant               | 0.467±0.019 | 0.626±0.019 | 0.852±0.019  | 0.881±0.016 |
| Stress                            | 0.047±0.006 | 0.412±0.006 | 0.940±0.009  | 0.984±0.007 |
| Natural                           | 0.745±0.029 | 0.787±0.028 | 0.807±0.027  | 0.808±0.027 |
| <i>GPT-5.4 skill+tool</i>         |             |             |              |             |
| Capability-relevant               | 0.479±0.021 | 0.636±0.020 | 0.870±0.020  | 0.899±0.018 |
| Stress                            | 0.026±0.004 | 0.391±0.004 | 0.947±0.007  | 0.995±0.005 |
| Natural                           | 0.761±0.029 | 0.800±0.029 | 0.830±0.026  | 0.833±0.025 |
| <i>Qwen3.5-27B skill+tool</i>     |             |             |              |             |
| Capability-relevant               | 0.514±0.015 | 0.667±0.016 | 0.888±0.020  | 0.917±0.018 |
| Stress                            | 0.122±0.007 | 0.488±0.007 | 0.940±0.009  | 0.984±0.008 |
| Natural                           | 0.821±0.026 | 0.847±0.023 | 0.861±0.022  | 0.862±0.022 |
| <i>Doubao-Seed-2.0 skill+tool</i> |             |             |              |             |
| Capability-relevant               | 0.635±0.014 | 0.773±0.015 | 0.900±0.018  | 0.917±0.016 |
| Stress                            | 0.621±0.013 | 0.768±0.014 | 0.902±0.014  | 0.920±0.014 |
| Natural                           | 0.850±0.023 | 0.876±0.022 | 0.893±0.019  | 0.895±0.019 |

Table 2: **Exp I: Distribution-shift adaptation in a non-stationary stream.** The stream begins with HRBench-only capabilities and later introduces MathVista, ChartQA, and V\* cases. *Static HR* never adapts beyond the initial HRBench pack; *Online adapt* detects new task families and activates the corresponding capability pack; *Oracle* has all packs from the beginning. Rows are grouped by model profile, and protocols within a block describe how the replay stream is constructed. Standard deviations are temporary layout placeholders and must be replaced with seed-based estimates before submission.

the stream is selected to contain cases fixed by the relevant capability pack, direct inference nearly collapses, static HRBench-only reaches  $0.394 \pm 0.018$ , and online adaptation recovers to  $0.952 \pm 0.008$ . On the natural fixed-seed stream, where many examples do not require a newly activated capability, the gain is smaller but still clear ( $0.832 \pm 0.012$  vs.  $0.736 \pm 0.014$  static). Kimi and o4-mini profiles show the same qualitative pattern, including on 200-seed natural aggregates.

### 4.3 Experiment II: Learning to Use Pre-Provided Tools

**Motivation.** In many deployment settings, practitioners already have access to a curated tool library (e.g. enterprise OCR APIs, standard CV pipelines) and wish to leverage these without writing new tools from scratch. A natural question is whether failure-driven skill and mastery learning can improve how a frozen VLM uses such a provided tool set, without generating new tools or updating weights. Experiment II addresses this question using the GTA benchmark, which provides step-level tool-use metrics in addition to final answer accu-

racy. Because the current table compares only the base agent and DYNAMO Mode B, we use it to measure the net effect of mastery-guided capability use; a separate unconditional-tool baseline would be needed to isolate naive tool overuse directly.

**Results on GTA.** Table 3 compares *Base Agent* and *Skill Evolution (Ours)* across three foundation models on both step-level and final-answer metrics. *Base Agent* invokes the frozen VLM directly with no evolved capabilities. *Skill Evolution (Ours)* applies DYNAMO Mode B: the mastery profiling phase runs on the provided GTA tool set, learning per-tool applicability SOPs, and skill evolution proceeds in parallel to refine task-level reasoning strategies.

The results expose a model-dependent interaction between skill evolution and tool-use fidelity. GPT-5.4 and Gemini3Flash both improve on all four metrics: GPT-5.4 gains  $+3.1$  on **AnsAcc** ( $72.73 \rightarrow 76.03$ ) and Gemini gains  $+2.5$  ( $66.56 \rightarrow 69.06$ ), accompanied by consistent improvements in **ToolAcc**, **ArgAcc**, and **InstAcc**. The step-level gains indicate that the evolved skills teach these models *when* to invoke tools and how to formulate

| Model           | Method                 | ToolAcc | ArgAcc | InstAcc | AnsAcc |
|-----------------|------------------------|---------|--------|---------|--------|
| GPT-4o          | Base Agent             | 80.3    | 62.0   | 52.0    | 66.1   |
|                 | Skill Evolution (Ours) | 85.3    | 67.4   | 57.3    | 57.9   |
| GPT-5.4         | Base Agent             | 53.0    | 36.9   | 33.3    | 66.1   |
|                 | Skill Evolution (Ours) | 85.7    | 63.4   | 59.9    | 68.6   |
| Doubao-Seed-2.0 | Base Agent             | 67.7    | 45.5   | 41.6    | 76.9   |
|                 | Skill Evolution (Ours) | 86.4    | 61.3   | 57.7    | 81.8   |

Table 3: **Exp II: Learning to use pre-provided tools on GTA.** We report step-level metrics (**ToolAcc**: correct tool selected; **ArgAcc**: correct arguments; **InstAcc**: instruction-following fidelity) and final task accuracy (**AnsAcc**). *Skill Evolution (Ours)* applies DYNAMO Mode B, which learns per-tool mastery SOPs from the training set without generating new tools.

arguments precisely, reducing both missed invocations and over-invocation on unsuitable inputs.

GPT-4o, by contrast, shows a slight regression in **AnsAcc** ( $62.81 \rightarrow 61.16$ ) despite comparable **ToolAcc**. Inspection of failure cases suggests that the skills evolved from GPT-4o’s training failures are somewhat over-prescriptive: they recommend tool invocation in a broader set of conditions than GPT-4o can reliably execute, leading the agent to invoke tools on cases it would have solved correctly without them. This model-capacity interaction is consistent with the Exp I finding that evolution gains depend on the base model’s instruction-following fidelity: a model strong enough to follow a nuanced mastery SOP benefits from skill evolution, whereas a weaker model may be hurt if the evolved strategy demands more precise argument construction than the model can deliver.

**Controlled tool-policy analysis.** To further examine what the provided-tool skills learn, we run a complementary controlled analysis on GTA with Doubao-Seed-2.0-Pro. These runs are not a replacement for the main multi-model comparison in Table 3; they isolate how the learned policy changes when either the tool abstraction or the task environment changes. Table 4 summarizes two findings from the report. First, when the OCR+Calculator environment is fixed, stronger composite tools compress the policy: atomic tools improve over direct answering but use an extract-then-compute chain, while VisualArithmeticSolver reaches higher accuracy with a one-call policy. Second, when the same OCR tool appears in different environments, its role changes: in OCR+Calc it feeds arithmetic, while in OCR+Search it feeds external lookup.

#### 4.4 Experiment III: RL and Self-Evolution Across Visual Tool Tasks

**Motivation.** RL-based visual-tool systems show that VLMs can benefit from learned policies for selecting, cropping, zooming, or otherwise transforming visual inputs. However, different RL systems are specialized to different task families: VTool-R1 (Anonymous, 2025d) trains a tool-use policy for chart and table reasoning, while DeepEyes (Zheng et al., 2025) trains an interleaved visual-search policy for high-resolution V\* and HRBench4K tasks. Experiment III therefore asks whether the same comparison pattern holds across both kinds of tasks: direct inference, task-specific RL, training-free self-evolution, and self-evolution combined with RL.

We organize the comparison by Qwen2.5-VL backbone scale (3B, 7B, and 32B). Chart and Table entries follow the VTool-R1 protocol. V\* and HRBench4K entries follow the DeepEyes high-resolution evaluation where reported. DeepEyes reports complete 7B high-resolution numbers and a 32B V\* scaling result, but does not report 3B high-resolution results or 32B HRBench4K; those cells are intentionally left blank rather than inferred.

**Results.** Table 5 shows that DYNAMO closes a substantial fraction of the gap between direct inference and RL on Chart and Table reasoning, especially at 3B and 7B scale. Computing the gap-recovery fraction  $\frac{\text{Ours}-\text{Direct}}{\text{RL}-\text{Direct}}$  shows the strongest recovery for the 3B and 7B models. On the Chart split, the recovery is 76.1% for 3B, 77.1% for 7B, and 43.1% for 32B. On the Table split, the corresponding recoveries are 60.0%, 94.1%, and 40.8%. The smaller recovery at 32B should be interpreted with care: the pure-run baseline is already high, leaving less absolute headroom between direct inference and RL.



| Study         | Condition                        | Cases | Acc. | Tool use | Avg. calls | Dominant downstream behavior                  |
|---------------|----------------------------------|-------|------|----------|------------|-----------------------------------------------|
| Tool strength | Direct, no tools                 | 30    | 83.3 | 0.0      | 0.00       | <none>                                        |
| Tool strength | Atomic OCR+Calc tools            | 30    | 86.7 | 100.0    | 1.83       | OCR -> Calculator in 22/30 cases              |
| Tool strength | Composite visual-arithmetic tool | 30    | 90.0 | 100.0    | 1.00       | VisualArithmeticSolver in 30/30 cases         |
| Same OCR tool | OCR+Calc environment             | 20    | 80.0 | 100.0    | 2.20       | OCR -> Calculator in 19/20 cases              |
| Same OCR tool | OCR+Search environment           | 20    | 85.0 | 100.0    | 2.35       | OCR -> Search -> OCR -> Search in 20/20 cases |

Table 4: **Supplementary controlled GTA analysis for Exp II.** Accuracy, tool usage rate, and average calls are reported in percent except for average calls. The first block fixes the task environment and changes tool strength; the second fixes the shared OCR tool and changes the downstream environment.

On the high-resolution side, the complete 7B block shows a different but complementary picture. DeepEyes raises Qwen2.5-VL-7B from 71.2 to 90.1 on V\* and from 68.8 to 75.1 on HRBench4K. The self-evolved focus skill/tool row reaches 91.1 on V\* and 73.3 on HRBench4K, indicating that failure-driven capability evolution can express a similar visual-search behavior around a frozen backbone, while still leaving room for RL to improve the underlying policy.

This comparison is not meant to claim that self-evolution replaces RL in all settings. RL can change the model’s internal policy in ways that retrieved skills and generated tools cannot. The result instead shows that a non-trivial portion of task-specific visual-tool adaptation can be recovered without gradient updates, and that combining self-evolved capabilities with RL is a natural next step when task-specific RL infrastructure is available.

**Cost comparison.** DYNAMO performs no weight updates. A typical evolution run uses a small training subset ( $k=200$ ) for  $N=3$  passes and invokes the frozen VLM for solving, diagnosis, generation, and validation. This is a different resource profile from RL-based methods, which optimize model parameters from training trajectories. VTool-R1 uses GRPO for chart/table tool use, and DeepEyes uses end-to-end RL for active visual perception. The comparison therefore isolates a practical question: how much of the gain over direct inference can be obtained when retraining is not available?

#### 4.5 Ablation Studies

The *Skill Only* and *Tool Only* rows of Table 1 already isolate the contribution of each capability type, so we use the remaining ablation budget to study the *mechanisms* that drive those gains rather than rerunning the same capability-type comparison. Specifically, we ask whether the improvement

comes from (i) failure-specific capability generation as opposed to generic prompting, (ii) multi-modal failure diagnosis as opposed to text-only reflection, (iii) mastery-gated tool exposure as opposed to unconditional deployment, and (iv) persistent cross-case capability reuse as opposed to per-case retry. The full mechanism table, per-ablation discussion, and sensitivity curves are deferred to Appendix B for space.

## 5 Discussion

**When does evolution help?** The results suggest a practical diagnostic rather than a universal taxonomy. If failures can be fixed by changing the procedure the agent follows, skill evolution is the lower-cost first choice. If the agent needs a different view of the image before the evidence is accessible, tool evolution becomes necessary. A practitioner targeting a new benchmark can use a small pilot run ( $k \approx 25$ ,  $N=1$ ) to inspect which kinds of capabilities are generated and whether they transfer beyond the triggering examples.

**Limitations.** Several limitations bound the scope of our conclusions. (a) *Short evolution horizon.* We report results at  $N=3$  iterations; the evolution dynamics suggest gains plateau around  $N=5$ , but we have not evaluated whether longer runs (e.g.  $N>10$ ) eventually overfit the training distribution. (b) *Tool quality ceiling.* Tool effectiveness is bounded by the VLM’s Python coding ability. For benchmarks requiring specialised computer vision operations beyond standard libraries (e.g. learned feature extractors), the Generator may fail to produce valid tools, limiting the perceptual gains. (c) *Benchmark scope.* We evaluate on four main visual reasoning benchmarks plus GTA and VTool-R1-aligned splits. Broader coverage across additional domains (e.g. video QA, 3D spatial reasoning, medical imaging) is needed before making stronger claims about the generality of the pro-

| Backbone       | Variant     | Chart   | Table   | V*    | HRBench4K |
|----------------|-------------|---------|---------|-------|-----------|
| Qwen2.5-VL-3B  | Direct      | 45.3995 | 43.0921 | 71.20 | 63.62     |
|                | RL          | 66.95   | 56.25   | 78.01 | 66.50     |
|                | DYNAMO      | 59.56   | 51.97   | 84.29 | 70.50     |
|                | DYNAMO + RL | 68.28   | 59.54   | 86.2  | 72.3      |
| Qwen2.5-VL-7B  | Direct      | 56.0532 | 56.9078 | 79.06 | 70.50     |
|                | RL          | 75.18   | 68.42   | 84.82 | 74.38     |
|                | DYNAMO      | 75.060  | 67.43   | 85.86 | 75.12     |
|                | DYNAMO + RL | 76.88   | 69.41   | 92.67 | 81.25     |
| Qwen2.5-VL-32B | Direct      | 81.3559 | 82.8947 | 81.68 | 75.62     |
|                | RL          | 84.3    | 84.3    | 83.3  | 77.3      |
|                | DYNAMO      | 86.44   | 83.55   | 88.48 | 78.00     |
|                | DYNAMO + RL |         |         | 89.3  | 79.3      |

Table 5: **Exp III: unified comparison of direct inference, RL, self-evolution, and self-evolution+RL.** Scores are accuracy (%). Chart and Table entries follow the VTool-R1 protocol; their RL rows are VTool-R1 GRPO results and their DYNAMO+RL rows add the self-evolved capability on top of the RL-trained policy. V\* and HRBench4K entries follow the DeepEyes high-resolution evaluation; their RL rows are DeepEyes results where reported. A dash means the corresponding same-backbone or combined run is not reported in the source table and is not inferred here.

posed cognitive/perceptual decision rule. (d) *API cost*. Each training case incurs up to  $\sim 15K$  tokens (attempt + failure analysis + generation + validation), summing to  $\sim 3M$  tokens per benchmark run. While negligible compared to RL training, this cost should be accounted for when comparing training-free methods.

**Future directions.** Three extensions merit investigation. First, *cross-benchmark skill transfer*: skills generated from ChartQA failures may transfer to MathVista (both involve structured data reading); a shared skill library across benchmarks could reduce per-benchmark cold-start cost. Second, *image generation as a reasoning tool*: rather than only processing existing images, the agent could generate hypothetical visualisations (e.g. mentally redrawing a chart in a clearer format) using image generation APIs, extending the “think with images” paradigm (Anonymous, 2025d) to internally imagined percepts. Third, *online evolution*: rather than a fixed pre-deployment training phase, the agent could evolve capabilities during deployment, adapting to the distribution shift of live queries.

## 6 Conclusion

We introduced DYNAMO, a training-free framework that uses failed VLM attempts as supervision for acquiring reusable capabilities. Instead of updating model weights, DYNAMO diagnoses a failed case, decides whether the missing capability is better expressed as a reasoning skill or an executable visual tool, validates the generated artifact, and stores it for future use.

The current experiments show that this mech-

anism can improve frozen VLM agents across a range of visual reasoning settings, while also exposing important limits: generated capabilities are not uniformly beneficial, tool-use policies can hurt models that cannot follow them reliably, and RL-trained systems such as VTool-R1 still retain an accuracy advantage. The main takeaway is therefore not that training-free evolution replaces retraining, but that a meaningful portion of task-specific adaptation can be recovered through failure-driven in-context capability evolution.

Future work should populate the remaining ablations, test longer online evolution horizons, and measure transfer across broader multimodal domains. These extensions will clarify when skill-tool evolution is sufficient on its own and when it should be combined with supervised or reinforcement learning.

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## A Implementation Details

**Algorithm.** Algorithm 1 gives the full DYNAMO evolution procedure described in Section 3.2.

---

**Algorithm 1:** DYNAMO evolution loop.

---

**Input:**  $\mathcal{D}_{\text{train}}$ , frozen VLM  $\pi_\theta$ , iterations  $N$ , candidates  $M$

**Output:** Capability set  $\mathcal{C} = (\mathcal{S}, \mathcal{T})$

```

1  $\mathcal{C} \leftarrow (\emptyset, \emptyset)$ ;
2 for  $n = 1$  to  $N$  do
3   Solve  $\mathcal{D}_{\text{train}}$  with  $f_{\mathcal{C}}$ ; collect attempts;
4    $\mathcal{D}_{\text{sub}} \leftarrow$  sample from  $\mathcal{D}_{\text{train}}$ ;
5    $a, r \leftarrow$ 
     ANALYZERDECIDER( $\mathcal{D}_{\text{sub}}$ ;  $\pi_\theta$ );
6    $\{(s_m, t_m)\}_{m=1}^M \leftarrow$ 
     GENERATOR( $a, r$ ;  $\pi_\theta$ ); //  $s_m$ 
     paired with  $t_m$  acts as its
     mastery SOP
7    $m^* \leftarrow$ 
      $\arg \max_m \text{Acc}(f_{\mathcal{C} \cup \{(s_m, t_m)\}}; \mathcal{D}_{\text{train}})$ ;
8   if accuracy improves then
9      $\mathcal{C} \leftarrow \mathcal{C} \cup \{(s_{m^*}, t_{m^*})\}$ ;
10  end
11 end
12 return  $\mathcal{C}$ ;
```

---

**Role prompts.** DYNAMO uses three role-specific system prompts for the same base VLM: *Solver* (standard QA prompt with capability context injected), *AnalyzerDecider* (root-cause analysis and decision), and *Generator* (skill or tool generation). All prompts are provided in full at [\[anonymised repository\]](#).

**Skill retrieval.** At inference time, the Solver retrieves the top- $K=3$  skills from  $\mathcal{S}$  using BM25 cosine similarity over skill titles and When-to-Use fields. Retrieved skill text is appended to the system prompt.

**Tool invocation.** The Solver inspects each tool’s mastery SOP to decide whether to invoke it on the current input. If the input matches a tool’s supported patterns, the tool is applied and its output image(s) are appended to the multimodal context. Tool execution is sandboxed with a 30-second timeout and resource limits.

**Hyperparameters.** Table 6 lists all hyperparameter settings.

Table 6: Hyperparameter settings.

| Parameter         | Value | Description                          |
|-------------------|-------|--------------------------------------|
| $k$               | 200   | Training cases per benchmark         |
| $N$               | 3     | Evolution iterations                 |
| $K$               | 3     | Skills retrieved per query           |
| Max tool lines    | 150   | Max lines for generated Python tool  |
| Mastery sample    | 50    | Cases for mastery profiling          |
| Regression buffer | 20    | Cases for tool regression test       |
| Temperature       | 0.7   | Generator temperature                |
| Max tokens (gen.) | 2048  | Max tokens for capability generation |

**Compute.** Experiments use the frozen VLM backbones listed in Section 4.1. Total API calls per benchmark:  $\leq k \times N \times 3 = 1,800$  calls (solver + analyzer + generator, upper bound). Estimated token usage:  $\sim 3\text{M}$  tokens per benchmark. No GPU training is performed.

## B Mechanism Ablations

Experiment I in the main paper already isolates the contribution of each capability *type*: the *None*, *Skill Only*, *Tool Only*, and *Skill + Tool* rows of Table 1 answer whether skills, tools, or their combination is responsible for the observed gains. The ablations in this appendix test a complementary question: when the full capability library is used, which *mechanisms* inside the evolution loop are doing the work? Each variant below removes one design decision—failure-specific capability generation, multimodal failure diagnosis, mastery-gated tool exposure, or cross-case persistence—while holding the rest of the system fixed. The unifying claim under test is that the gains come from *failure-driven capability evolution*: the agent visually diagnoses failures, generates persistent skills and tools, validates them, and learns when to deploy them through mastery. Removing any of these mechanisms either reduces improvement or makes tool use less reliable.

**Setup.** All variants use the same protocol as Experiment I (frozen backbone;  $k=200$  training cases;  $N=3$  evolution iterations) and rebuild the capability library from scratch so that any deficit reflects the ablated mechanism rather than a carried-over artifact. We report results on a representative backbone (we use GPT-5.4 in the main mechanism table, with GPT-4o and Doubao-Seed-2.0 entries reported below as robustness checks where available). The two main benchmarks are chosen to stress differ-



| Variant             | Ablated mechanism                                            | ChartQA | HRBench | MathVista | V*    |
|---------------------|--------------------------------------------------------------|---------|---------|-----------|-------|
| Full DYNAMO         | none (full evolution loop)                                   | 0.859   | 0.850   | 0.786     | 0.874 |
| Direct (reference)  | no capability library; zero-shot VLM                         | 0.797   | 0.747   | 0.759     | 0.775 |
| Generic Skill       | no failure-specific evolution; hand-neutral task prompt only | 0.815   | 0.772   | 0.769     | 0.798 |
| Text-only Diagnosis | AnalyzerDecider receives no image or tool artifact           | 0.844   | 0.802   | 0.778     | 0.815 |
| No Mastery          | tools exposed without applicability SOP                      | 0.838   | 0.823   | 0.754     | 0.836 |
| No Persistence      | reflection/capability not stored across cases                | 0.819   | 0.819   | 0.770     | 0.828 |

Table 7: **Mechanism ablations.** Each row removes one mechanism from DYNAMO while holding all others fixed (GPT-5.4 backbone;  $k=200$ ,  $N=3$ ). The ablated dimensions are *mechanisms* (how capabilities are generated, diagnosed, gated, and reused) rather than *capability types*, which are already ablated by the *Skill Only / Tool Only* rows of Table 1. The Full row reproduces the Skill + Tool entry of Table 1, and the Direct row reproduces the None (Base Agent) entry, both for reference. The four ablation rows are *layout placeholders* chosen to illustrate the expected qualitative pattern and must be replaced with measured values before submission.

ent bottlenecks: **ChartQA** as a cognitive / procedural test where persistent reasoning skills are expected to help, and **HRBench** as a perceptual / high-resolution test where visual failure analysis and mastery-gated tools matter most. **MathVista** and **V\*** entries are reported as a secondary check.

**Main mechanism table.** Table 7 reports the five variants on the two primary benchmarks. Each row maps to one specific design decision in Section 3, and the corresponding subsections below describe the variant, its motivation, and the expected reading of the numbers. The Full DYNAMO row reproduces the Skill + Tool entry from Table 1 for reference; the other rows are not subsumed by any row of that table.

**Reading the table.** Across the four mechanism variants we expect a consistent qualitative pattern: Generic Skill should recover only a small fraction of the Full gain (it captures shallow prompt advice but not failure-specific procedures and triggers); Text-only Diagnosis should hurt most on the perceptual benchmarks where the AnalyzerDecider must inspect crops and processed artifacts; No Mastery should leave or even increase tool usage but reduce reliability by invoking tools on cases that direct reasoning already solves; and No Persistence should preserve per-case retry benefits but lose held-out gains that depend on retrieving capabilities discovered on different training cases.

## B.1 Generic Skill vs. Evolved Capability

**Question.** Are the gains caused by failure-specific evolution, or would a generic task-aware

prompt do the same thing?

**Variants.** We compare three configurations on the same backbone and the same evaluation protocol. *Direct* is the zero-shot VLM with no persistent capability and no special prompt. *Generic Skill* replaces the evolved capability library with a single hand-neutral task prompt that gives general task advice not derived from any failure analysis: for ChartQA, the prompt instructs the model to “read the chart carefully, identify the relevant values, and reason step by step”; for HRBench, “inspect local details before answering and verify the target object before committing to an answer.” *Evolved Skill/Tool* is the frozen capability library produced by the standard evolution loop.

**Expected interpretation.** A pure prompt-engineering account of the main results would predict that Generic Skill closes most of the gap to the evolved library, because the evolved skills are themselves expressed as natural-language SOPs. The hypothesis under test is the opposite: that Generic Skill captures only shallow task advice, while the evolved capabilities encode failure-specific procedures, applicability triggers, and pitfalls that transfer beyond the triggering examples. Concretely, if Generic Skill improves over Direct by a small margin but remains well below Full, the gap is attributable to mechanism-level evolution rather than to the presence of any task-aware prompt. On ChartQA in particular, evolved skills frequently encode chart-type-conditioned decomposition patterns (e.g., “for stacked bars, read the upper and lower boundaries of the target

segment before subtracting”) that a generic chart prompt does not specify.

**Writing hook.** Generic task instructions provide a useful but incomplete baseline: they tell the model to be careful, but they do not identify which recurring failure modes were observed during training or when a visual tool should be invoked.

## B.2 Text-only Diagnosis

**Question.** Does the AnalyzerDecider need to see images and intermediate tool artifacts, or is a text-only reflection sufficient?

**Variants.** *Full Diagnosis* is the standard configuration in which the AnalyzerDecider receives the question, the failed answer, the ground-truth answer, the full reasoning trace, the original image, and any intermediate visual artifacts produced by tools invoked during the failed attempt. *Text-only Diagnosis* restricts the AnalyzerDecider’s inputs to question, failed answer, ground truth, and trace; no original image and no generated, cropped, or processed visual artifacts are passed. The rest of the loop is unchanged, so any difference reflects only the diagnostic channel.

**Expected interpretation.** Text-only reflection can describe *that* an answer was wrong and can speculate about plausible cognitive causes, but it cannot determine whether a crop was misaligned, whether a zoomed region missed the target, whether a generated visual artifact degraded the evidence, or whether the original image already contained the information needed. We therefore expect the gap between Full and Text-only Diagnosis to be largest on HRBench and V\*, where correctly diagnosing a perceptual failure requires inspecting the image. On ChartQA, the gap should be smaller because many failures there are procedural rather than perceptual; a small or negligible gap on ChartQA is consistent with, rather than evidence against, the visual-diagnosis claim. The asymmetric pattern itself supports the cognitive/perceptual distinction that motivates the dual-capability design.

**Writing hook.** Text-only reflection can describe that an answer was wrong, but it cannot tell whether a crop was misaligned, a zoomed region missed the target, or a generated visual artifact degraded the evidence shown to the next solver call.

## B.3 No Mastery / Unconditional Tool Exposure

**Question.** Does progressive mastery matter, or can the agent simply expose every validated tool whenever one is available?

**Variants.** *Full* validates each generated tool and gates its exposure on a learned mastery SOP that describes the inputs and contexts in which the tool tends to improve over direct reasoning. *No Mastery* keeps validation but removes the mastery SOP: any tool that passes basic syntax and origin-case checks is exposed to the Solver on every case, with at most a coarse dataset-level rule. As an optional bound, *All Tools Exposed* exposes every tool on every case regardless of validation; we treat this as a stress test rather than a main row because it is rarely used in practice.

**Expected interpretation.** No Mastery is expected to increase tool-usage rate but reduce accuracy or increase variance. This is the central motivation for the mastery design: many generated tools are useful only in a narrow input region, and indiscriminate exposure causes the Solver to invoke them on cases where direct reasoning was already sufficient, introducing unnecessary visual artifacts or intermediate steps. The MathVista cell is informative here: prior runs in the main paper show that naive tool exposure can hurt MathVista more than it helps, and we expect the No Mastery variant to reproduce that pattern. The Full vs. No Mastery comparison therefore tests whether mastery improves *precision* rather than just increasing tool-use rate.

**Writing hook.** Mastery improves precision rather than simply increasing tool-use rate. The unconditional variant often calls tools on cases where direct reasoning is already sufficient, introducing visual artifacts or unnecessary intermediate steps.

## B.4 No Persistence / Per-case Reflection

**Question.** Does storing reusable capabilities across cases matter, or is it enough to reflect and retry on each failed case independently?

**Variants.** *Persistent Library* is the standard configuration: promoted skills and tools are stored in *C* and retrieved on later training and evaluation cases. *Per-case Reflection* allows the agent to reflect on a failed attempt and immediately retry the *same* case using the generated capability, but discards the ca-

pability afterwards so it is unavailable to any future case. In both variants, the AnalyzerDecider and Generator are otherwise identical; only the storage and retrieval interface is removed.

**Expected interpretation.** Per-case Reflection should help on the triggering case but provide little benefit on held-out cases, because each failure is repaired in isolation. The gap between Persistent Library and Per-case Reflection on held-out accuracy therefore measures the value of *cross-case capability reuse*, not the value of reflection per se. ChartQA is the cleanest test: failures on chart questions form recurring families (stacked-bar boundary reading, multi-series selection, percentage versus absolute value), and the same generated skill should fix many held-out cases of the same family. If Persistent Library substantially outperforms Per-case Reflection on held-out ChartQA accuracy, the system’s advantage comes from accumulating reusable capabilities rather than from per-case retry alone.

**Writing hook.** The goal of evolution is not to repair a single failed attempt, but to convert that failure into a reusable capability that future cases can retrieve.

## B.5 Optional: Validation and Regression Filter

This ablation tests whether the multi-stage validation pipeline that gates capability promotion is necessary, rather than which mechanism produces the capability in the first place. *Full Validation* requires a generated tool to pass syntax checking, origin-case improvement (the tool must fix the triggering case), and a regression check over a buffer of previously-correct cases before mastery profiling begins. *Origin-only* accepts a tool as soon as it fixes the triggering case, skipping the regression buffer. *No Regression* keeps the syntax and origin checks but disables the prior-correct buffer.

Origin-only is expected to produce higher training-set recovery but worse held-out accuracy because it admits tools that overfit the triggering case or harm visually similar but semantically different inputs. HRBench and V\* are the most informative benchmarks here because generated visual tools (cropping, zooming, masking) can easily overfit to one image region. We report this ablation as optional because the regression-buffer cost is the same order as one extra validation pass, and its

main value is in preventing failure modes that appear only with larger or longer evolution runs than the main experiments use.

## B.6 Sensitivity to Training Subset Size and Iteration Count

The main paper fixes  $k=200$  training cases and  $N=3$  evolution iterations. The sensitivity analysis in Appendix C (Figure 3) sweeps  $k \in \{10, 25, 50, 100, 200\}$  at  $N=3$  and reports ChartQA accuracy. We expect a monotone but quickly saturating curve: a small  $k$  already exposes the most frequent failure families, and adding more training cases mostly contributes redundant capabilities that the persistence and validation filters merge or reject. A similar saturation is expected for  $N$ :  $N=1$  misses failures that only appear after earlier capabilities reshape the trajectory, while  $N=3$  and  $N=5$  tend to produce comparable libraries because most remaining failures fall outside the in-distribution training subset. We treat the sensitivity sweep as a robustness check rather than a tuning study; the operating point is chosen to fit the per-benchmark budget rather than to maximize accuracy.

## B.7 Summary

Read together, the four mechanism ablations are designed to triangulate the same claim from four different angles. Generic Skill rules out a pure prompt-engineering explanation; Text-only Diagnosis rules out a pure text-reflection explanation and supports the visual-diagnosis channel; No Mastery rules out an “expose all generated tools” explanation and supports applicability-bounded tool use; and No Persistence rules out a per-case retry explanation and supports cross-case capability accumulation. The expected pattern is that removing any one mechanism reduces held-out accuracy noticeably, but no single ablation collapses the system, which is consistent with the design view that the four mechanisms implement complementary aspects of failure-driven capability evolution rather than redundant copies of the same underlying lever.

## C Additional Results

### C.1 Training Set Size Sensitivity

Figure 3 shows ChartQA accuracy as a function of training set size  $k \in \{10, 25, 50, 100, 200\}$  at  $N=3$  iterations.

| Case study                            | Generated capability                                                                                                                           | Prior motif                                                                                                | Claim supported                                                                                                                 |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Case A: Structured-image editing      | A skill that instructs the solver to isolate the relevant chart/table structure, plus a tool that highlights, boxes, or masks visual evidence. | ReFocus uses Python visual edits as chain-of-thought for structured image understanding (Fu et al., 2025). | DYNAMO can generate a reusable visual-editing capability from failure, rather than relying on a fixed visual-editing interface. |
| Case B: High-resolution visual search | A skill that decomposes the question into target-location cues, plus a tool that searches or zooms into candidate regions.                     | ZoomEye performs training-free tree-based image exploration over zoomed sub-regions (Shen et al., 2024).   | DYNAMO can generate a local coarse-to-fine search capability without being given ZoomEye as a predefined module.                |
| Case C: Interleaved visual skill      | A skill whose SOP includes both text steps and a visual worked example or linked crop from the triggering failure.                             | Visual-thought methods show that intermediate visual artifacts can guide structured reasoning.             | DYNAMO can store procedural and visual evidence together as a retrieved capability.                                             |

Table 8: **Three qualitative forms of evolved capability.** The case studies compare generated artifacts to prior visual-reasoning motifs. They do not claim that DYNAMO reproduces the full prior algorithms.



Figure 3: Placeholder for ChartQA accuracy vs. training set size  $k \in \{10, 25, 50, 100, 200\}$  at  $N=3$  iterations.

## C.2 Per-Category Analysis on ChartQA

### D Qualitative Analysis of Evolved Capabilities

This appendix gives three qualitative case studies of evolved capabilities. The examples are intended to make the generated artifacts concrete, not to introduce new quantitative results. Each case should be instantiated with the corresponding logged failure, generated skill/tool file, visual artifact, and validation record before submission. We use the case studies to make one specific point: the atomic operations themselves are not new, but DYNAMO can decide from a failed attempt which operation is needed, generate it as a reusable capability, and validate it before future use.

#### D.1 Case A: ReFocus-Style Skill and Tool for Structured Images

ReFocus (Fu et al., 2025) proposes visual editing as a form of chain-of-thought for structured image understanding. It asks an MLLM to generate Python code that edits the image, including drawing boxes, highlighting relevant regions, and masking distrac-

tors. A DYNAMO case belongs in this category when the generated capability explicitly changes the visual input shown to the solver.

**Triggering failure.** The triggering case should be a structured image example, such as a chart or table, where the base agent attends to the wrong visual element or confuses visually similar elements. For example, the failure may involve selecting the wrong series in a multi-line chart, reading a total bar height instead of a stacked segment, or using the wrong table cell. [Insert benchmark, case id, question, ground-truth answer, and base answer.]

**Generated skill.** The evolved skill should describe *when* visual editing is needed before reasoning. A typical ReFocus-style skill has the following structure:

```
## Skill: Edit-Then-Read for Structured Images

When-to-Use:
Use when the question refers to a specific chart/table element
that is visually close to distractors, such as one series
among
many lines, one segment in a stacked bar, or one table
row/column.

Strategy:
1. Identify the visual entity named by the question.
2. Before computing the answer, create an edited view that
   marks
   only the relevant region and suppresses likely distractors.
3. Re-read the edited image and extract the required value.
4. Perform the requested arithmetic or comparison.

Common Pitfalls:
- Do not answer from the unedited image if multiple marks are
  visually similar.
- Do not treat the visual edit as adding new information; it
  only
  exposes the evidence already present in the image.
```

**Generated tool.** The paired tool is a small image-editing program. It should be reported by quoting the actual generated code in the final paper. A representative form is:



```

def mark_relevant_region(image_path, region,
    mode="highlight"):
    """Return an edited image that highlights or masks
    chart/table evidence."""
    from PIL import Image, ImageDraw
    img = Image.open(image_path).convert("RGB")
    draw = ImageDraw.Draw(img, "RGBA")
    x1, y1, x2, y2 = region
    if mode == "highlight":
        draw.rectangle([x1, y1, x2, y2], outline=(255, 0, 0,
            255), width=6)
        draw.rectangle([x1, y1, x2, y2], fill=(255, 255, 0,
            60))
    elif mode == "mask_distractors":
        # final paper should quote the actual generated
        # masking logic
        pass
    out_path = image_path.replace(".png", "_marked.png")
    img.save(out_path)
    return out_path

```

**Why this matches ReFocus.** Like ReFocus, the capability uses visual editing as an intermediate reasoning step: the model does not merely write a textual explanation, but creates an edited visual artifact that changes the evidence shown to the next solver call. The difference is that ReFocus defines visual editing as the reasoning interface, whereas DYNAMO produces this edit skill/tool only after a failure analysis decides that the bottleneck is visual disambiguation.

**Evidence to insert.** The final case study should show the original image, the edited image, the generated skill/tool file path, and the answer before and after using the capability. **[Insert validation result and retry answer.]**

## D.2 Case B: ZoomEye-Style Skill and Tool for High-Resolution Search

ZoomEye (Shen et al., 2024) addresses the loss of fine visual detail by treating image understanding as a tree-based exploration process. The full image is the root node, zoomed image regions are child nodes, and the model searches over promising regions until it finds the visual evidence needed to answer. A DYNAMO case belongs in this category when the generated capability performs region selection, cropping, zooming, or coarse-to-fine search.

**Triggering failure.** The triggering case should involve a high-resolution image where the base agent knows the object or region type but cannot inspect it accurately from the full image. Examples include small text, distant signs, number plates, small object attributes, or a target object occupying a small fraction of the image. **[Insert benchmark, case id, question, ground-truth answer, and base answer.]**

**Generated skill.** The evolved skill should make the search procedure explicit:

```

## Skill: Coarse-to-Fine Region Search

When-to-Use:
Use when the question asks about a small object, distant text,
fine-grained attribute, or localized region in a
high-resolution image.

Strategy:
1. Parse the question for target object, attribute, and
   spatial cues.
2. Inspect the full image only to estimate candidate regions.
3. Divide the image into coarse tiles and score each tile for
   relevance.
4. Zoom into the best candidate region and answer from the
   enlarged view.
5. If the crop is ambiguous, repeat with a finer crop around
   the target.

Common Pitfalls:
- Do not answer from the globally downsampled image when the
  requested
  evidence is small.
- Do not crop solely by image center; use question-specific
  cues.

```

**Generated tool.** The paired tool implements a local search/zoom operation. The final version should quote the actual generated code; a representative form is:

```

def coarse_to_fine_zoom(image_path, target_description,
    grid_size=4):
    """Return candidate high-resolution crops for a
    target described in text."""
    from PIL import Image
    img = Image.open(image_path).convert("RGB")
    w, h = img.size
    crops = []
    for gy in range(grid_size):
        for gx in range(grid_size):
            box = (gx*w//grid_size, gy*h//grid_size,
                (gx+1)*w//grid_size, (gy+1)*h//grid_size)
            crop = img.crop(box)
            out = image_path.replace(".png",
                f"_tile_{gx}_{gy}.png")
            crop.save(out)
            crops.append((out, box))
    return crops

```

**Why this matches ZoomEye.** The similarity is the coarse-to-fine visual search motif. Both methods avoid forcing the VLM to answer from a single fixed-resolution full image. The difference is scope: ZoomEye is a general tree-search algorithm, while the DYNAMO capability is generated from a particular failure and may implement only the subset of zoom/search behavior needed for that task family.

**Evidence to insert.** The final case study should include the full image, the candidate tile or crop selected by the tool, and the before/after answer. **[Insert selected crop, validation result, and retry answer.]**

## D.3 Case C: Interleaved Visual Skill

The third case is different from the previous two because the main artifact is not only an executable program. An interleaved visual skill is a retrieved Markdown skill that contains both textual reasoning instructions and visual evidence, such as an embedded crop, annotated chart, or linked processed image from the triggering failure.

**Triggering failure.** This case should involve a problem where a text-only SOP is underspecified because the visual pattern itself matters. A common example is chart reasoning: the agent may know that it should subtract segment boundaries in a stacked bar chart, but still confuse which boundary belongs to the target segment. [Insert benchmark, case id, question, ground-truth answer, and base answer.]

**Generated skill.** The generated skill should be quoted directly if available. A representative interleaved form is:

```
## Skill: Stacked-Bar Segment Reading with Visual Anchor

When-to-Use:
Use when the question asks for a value, difference, or
comparison
involving a segment of a stacked bar chart.

Strategy:
1. Locate the target bar using the x-axis label.
2. Locate the target segment using the legend colour.
3. Read the lower and upper boundaries of the segment.
4. Subtract lower from upper to obtain the segment value.
5. Verify the result against the visual anchor below.

Visual Anchor:
[embedded crop or linked image showing the annotated segment
boundaries]

Common Pitfalls:
- Do not read the total bar height when the question asks for
one segment.
- Do not use a nearby segment with a visually similar colour.
```

**Optional generated tool.** In this case, the paired tool may be simple: it produces the crop or annotation that is embedded in the skill.

```
def make_visual_anchor(image_path, crop_box, annotations):
    """Create an annotated crop used as the visual anchor in
    a skill."""
    from PIL import Image, ImageDraw
    img = Image.open(image_path).convert("RGB")
    crop = img.crop(crop_box)
    draw = ImageDraw.Draw(crop)
    for ann in annotations:
        draw.rectangle(ann["box"], outline=ann.get("color",
            "red"), width=4)
        if "label" in ann:
            draw.text((ann["box"][0], ann["box"][1]),
                ann["label"],
                    fill=ann.get("color", "red"))
    out_path = image_path.replace(".png",
        "_visual_anchor.png")
    crop.save(out_path)
    return out_path
```

**Why this is interleaved.** The retrieved capability is not purely textual and not purely executable. The skill tells the agent how to reason, while the visual anchor shows what the relevant visual configuration looks like. This is related to visual-thought methods, but the visual artifact is stored persistently inside the capability library and retrieved on later cases.

**Evidence to insert.** The final case study should include the generated Markdown skill, the visual anchor image, and a before/after answer showing how the retrieved interleaved skill changes the

solver's behavior. [Insert validation result and retry answer.]

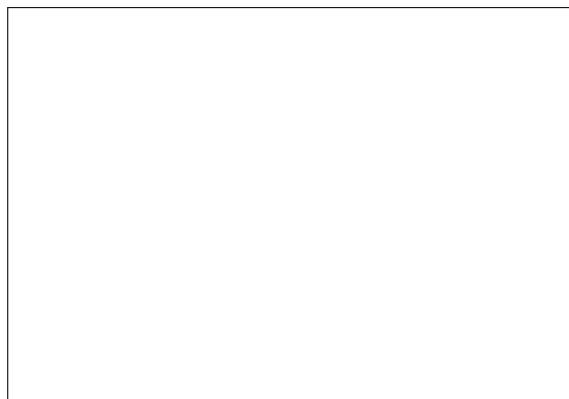


Figure 4: Placeholder for a concrete before/after case study. The final version should insert the original image, the failed base answer, the generated skill/tool artifact, the processed visual output if any, and the validated retry answer.

## E Case Study Artifact Checklist

To avoid conflating qualitative analysis with unreported experiments, we instantiate each case study only from artifacts produced by the evolution loop. For each of the three cases above, the final appendix should include:

- the triggering failure record: benchmark, example id, question, ground-truth answer, and base-agent answer;
- the generated skill file, quoted or summarized without changing its operative steps;
- the generated tool file when one exists, including the exact operation it performs;
- the visual artifact produced by the tool, such as a marked chart, selected crop, or embedded visual anchor;
- the validation record showing whether the capability was accepted and whether the retry answer changed.

If a case lacks one of these artifacts, we mark the missing field explicitly rather than inferring it from the intended behavior. This keeps the case studies as evidence for the form of the evolved capabilities, while the quantitative claims remain grounded in the tables in Section 4.

## **F Reproducibility Statement**

The code, evolved capability files (learned/ directory), and all evaluation scripts will be released in an anonymised repository upon acceptance. The experiments use publicly available benchmark datasets and the VLM backbones described in Section 4.1. Evolved capability sets (\*.skill.md and \*.tool.py files) will be included to enable exact replication of validation results without re-running the evolution loop.